

Bona and Tzvetkov [1] proved that the BBM equation is globally well - posed in $H^s(\mathbb{R})$ for $s \geq 0$. The same result also holds for the periodic case [2, 3]. This well - posedness is optimal, since if $s < 0$, the equation (1) is ill - posed in $H^s(\mathbb{R})$. In fact, Bona and Tzvetkov [1] also proved that for $s < 0$, the solution map $u_0 \mapsto u(t)$ of the equation (1) is not a C^2 map from $H^s(\mathbb{R})$ to $C([0, T]; H^s(\mathbb{R}))$.

Later, Panthee [4] improved this result by showing that the solution map is not continuous from $H^s(\mathbb{R})$ to $\mathcal{D}'(\mathbb{R})$ for any fixed $t > 0$ small enough.

Carvajal and Panthee [5] considered the following generalized BBM equation

$$\begin{cases} u_t + u_x + Lu_t + (u^{k+1})_x = 0, \\ u(0, x) = u_0(x), \end{cases} \quad (2)$$

where L is defined via the Fourier transform as $\widehat{Lu}(\xi) = |\xi|^{\alpha+1}\widehat{u}(\xi)$, with $\alpha > 0$.

Carvajal and Panthee [5] proved that the equation (2) is locally well - posed in $H^s(\mathbb{R})$ for some s depending on α and k . Moreover, if $s < \max\{0, \frac{1}{2} - \frac{\alpha}{k}\}$, the equation (2) is ill - posed in $H^s(\mathbb{R})$, which means that the solution map is not C^{k+1} - differentiable from $H^s(\mathbb{R})$ to $C([0, T]; H^s(\mathbb{R}))$ for any $T > 0$. When $\alpha = 1$, some global well - posedness results for (2) with periodic boundary conditions were obtained [3].

2 Model Derivation

From 2002 to 2004, Bona, Chen and Saut [6, 7] started from the free - surface Euler equations. Following the derivation context of the Boussinesq original equations, they considered the equivalent relationships among the physical quantities in the shallow - water region and combined the following expanded physical perturbation parameters

$$\alpha = \frac{A}{h_0} \ll 1, \quad \beta = \frac{h_0}{l^2} \ll 1, \quad S = \frac{\alpha}{\beta} = \frac{Al^2}{h_0^3} \approx 1,$$

where A and l represent the wave amplitude and wavelength in the classical sense, h_0 represents the undisturbed water depth, and S represents the Stokes number. The first - order form of the $abcd$ -system is

$$\begin{cases} \partial_t \eta + \partial_x w + \partial_x(w\eta) + a\partial_x^3 w - b\partial_x^2 \partial_t \eta = 0, \\ \partial_t w + \partial_x \eta + w\partial_x w + c\partial_x^3 \eta - d\partial_x^2 \partial_t w = 0. \end{cases} \quad (3)$$

Here, η is proportional to the deviation of the free - surface position at x ; $w = w(x, t)$ is proportional to the horizontal velocity at the point (x, z_0, t) at the depth z_0 in the flow domain. The constants a, b, c, d are not arbitrary, and they satisfy the relationships

$$\begin{cases} a = \frac{1}{2}\left(\theta^2 - \frac{1}{3}\right)\lambda, & b = \frac{1}{2}\left(\theta^2 - \frac{1}{3}\right)(1 - \lambda), \\ c = \frac{1}{2}(1 - \theta^2)\mu, & d = \frac{1}{2}(1 - \theta^2)(1 - \mu), \end{cases} \quad (4)$$